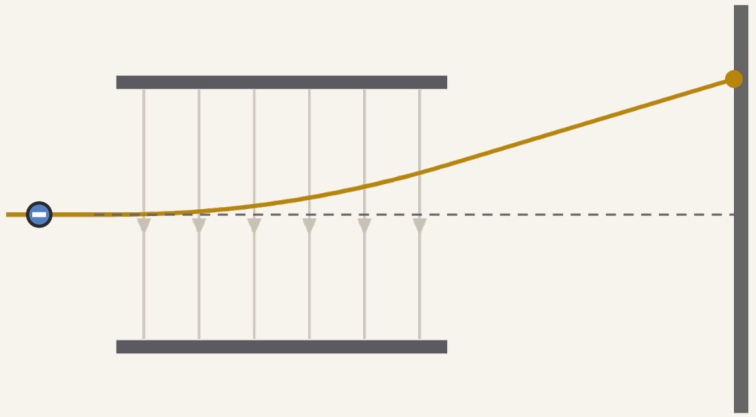
Electric Fields 2: Uniform Fields



Two plates, one steady field — enough to steer electrons.

Advanced Higher Physics

By the end of this deck…

1. I can **sketch** the field between parallel plates.
2. I can use $E = V/d$, $F = QE$ and $W = QV$.
3. I can use $QV = \frac{1}{2}mv^2$ and change **eV** to joules.
4. I can **explain** and **calculate** deflection between plates.
5. I can **describe** Millikan's method and find the charge on a drop.

Recap — from Higher

Between plates

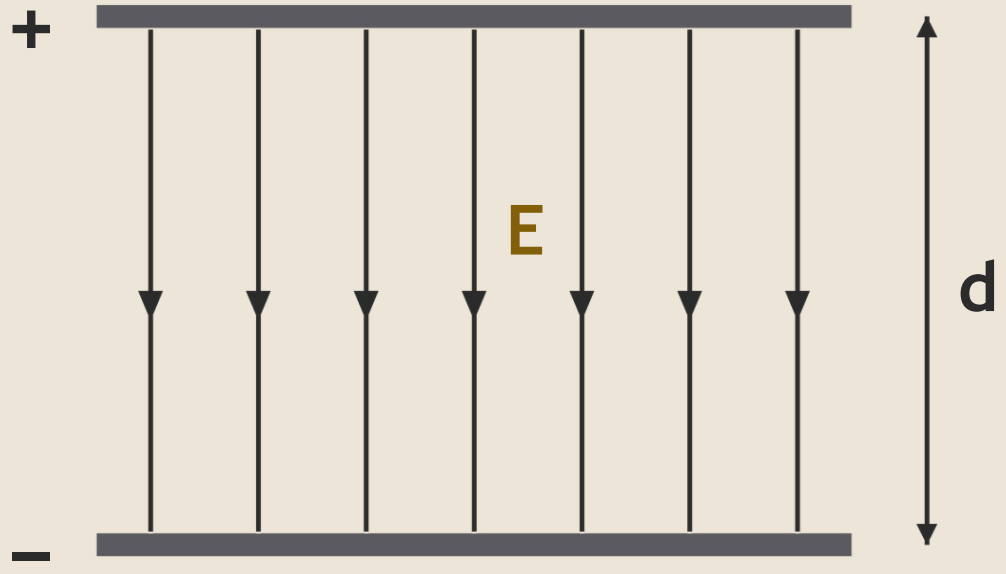
Lines parallel, + to −.

Work done

$W = QV$.

Speed

$E_k = QV$, so $\frac{1}{2}mv^2 = QV$.



2024 AH Physics: scan the code and answer these.

Q4(b)(i)

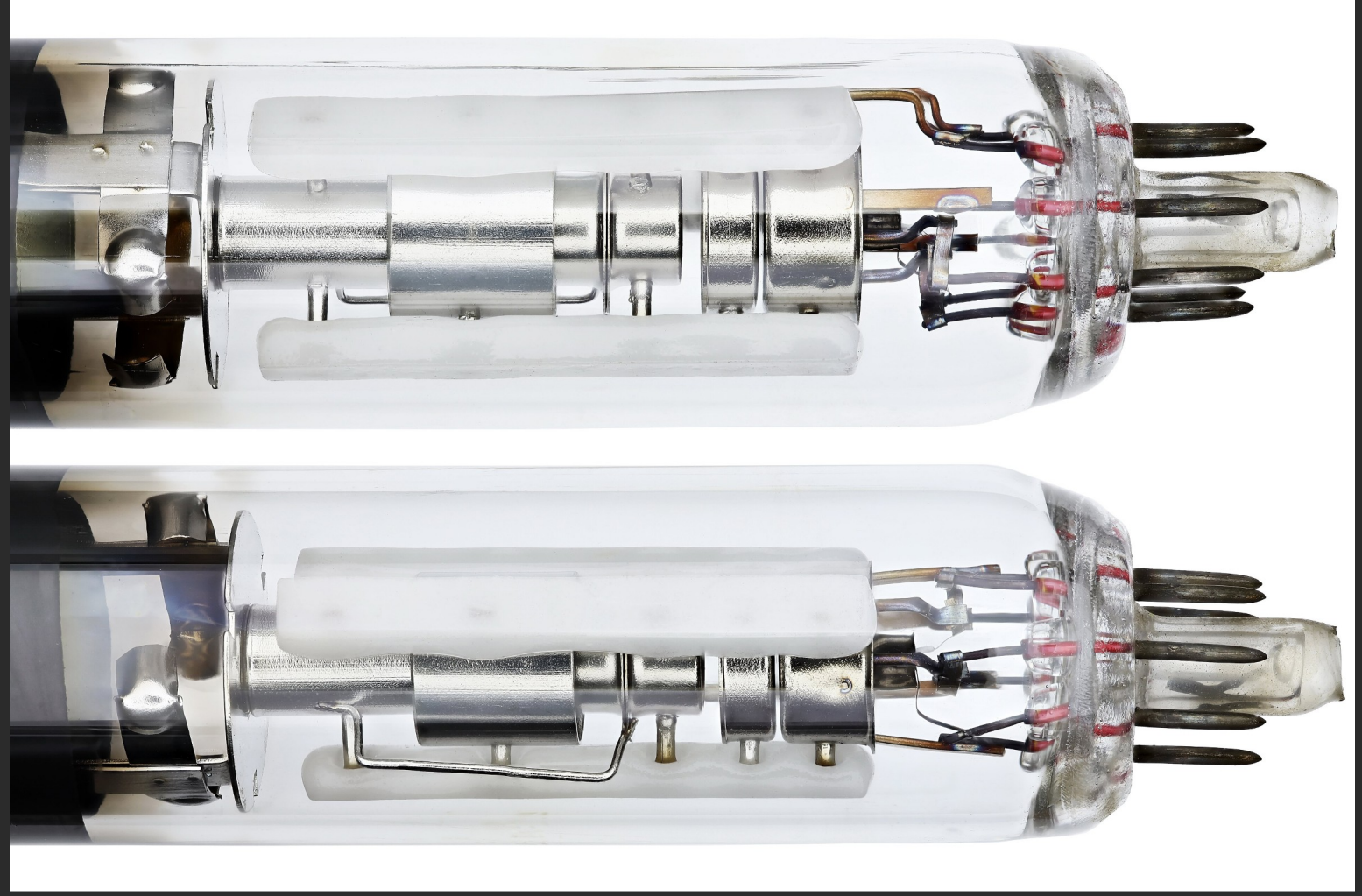
Q13(a)(ii)

Gravitation, then point charges.

Show every line of working.

Scan for the paper

Marking scheme



This electron gun came out of an old TV. How fast does it fire electrons?

PART 1

How fast can a field make an electron go?

Between parallel plates the force is the same everywhere.

By the end of this part…

1. I can **sketch** the field between parallel plates.
2. I can use $E = V/d$, $F = QE$ and $W = QV$.
3. I can use $QV = \frac{1}{2}mv^2$ for a charge accelerated from rest.
4. I can **state** what the **electronvolt** is.

The field between parallel plates

Uniform

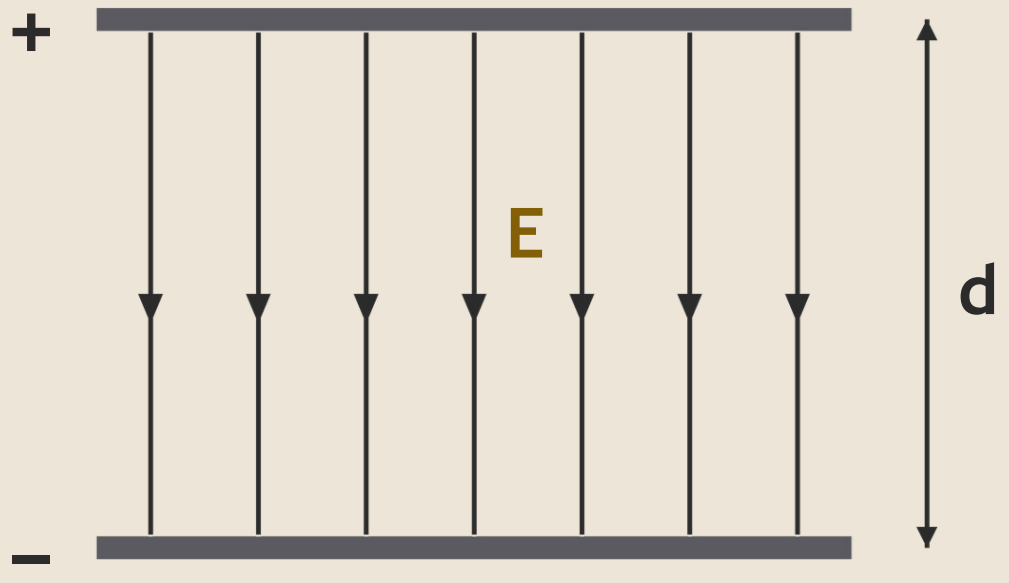
Same strength everywhere.

Lines

Straight, evenly spaced, + to −.

Edges

Ignore the bulge at the ends.



Example answer — 2022 Q14(b)

14. In a cathode ray oscilloscope, electrons are accelerated from rest between the cathode and anode. The electrons then travel with a constant horizontal velocity between the parallel deflection plates.

This arrangement is shown in Figure 14A.

(b) On the diagram below, **sketch** the electric field pattern between the parallel deflection plates. **1**

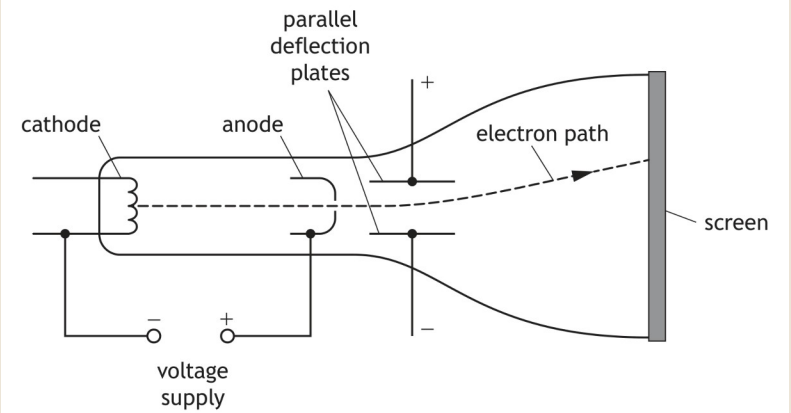
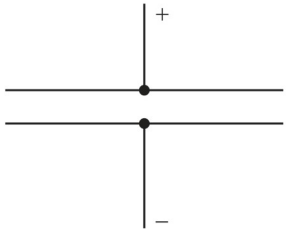


Figure 14A



Marking: 2022 Q14(b)

Question	Expected response	Max mark	Additional guidance
14 (b)	Parallel field lines from the + plate to the – plate, with arrows (1 mark)	1	Ignore end effects. Field lines must be straight/spaced uniformly. Field lines must start and end on the plates.

Field strength between the plates

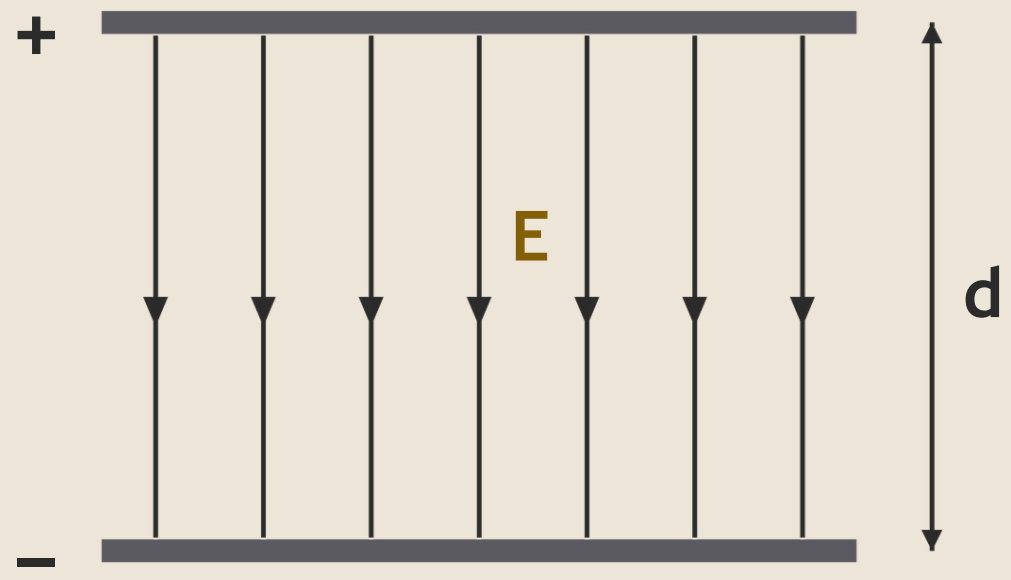
$$V = Ed$$

V: p.d. across the plates (V)

E: field strength (V m^{-1})

d: plate separation (m)

V m^{-1} is the same as N C^{-1}.



Force on a charge between the plates

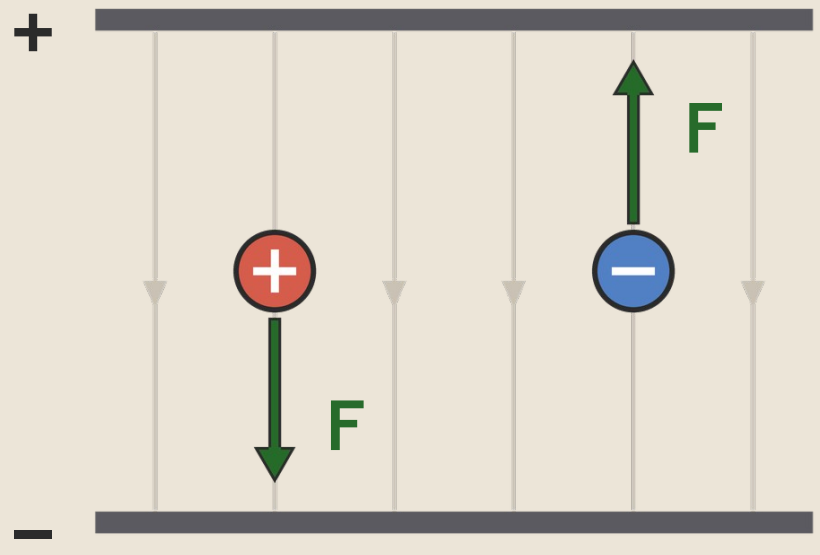
$$F = QE$$

F: force on the charge (N)

Q: the charge (C)

E: field strength (N C^{-1})

Same F anywhere between the plates.



+ pushed along E; − against it.

Work done moving a charge across

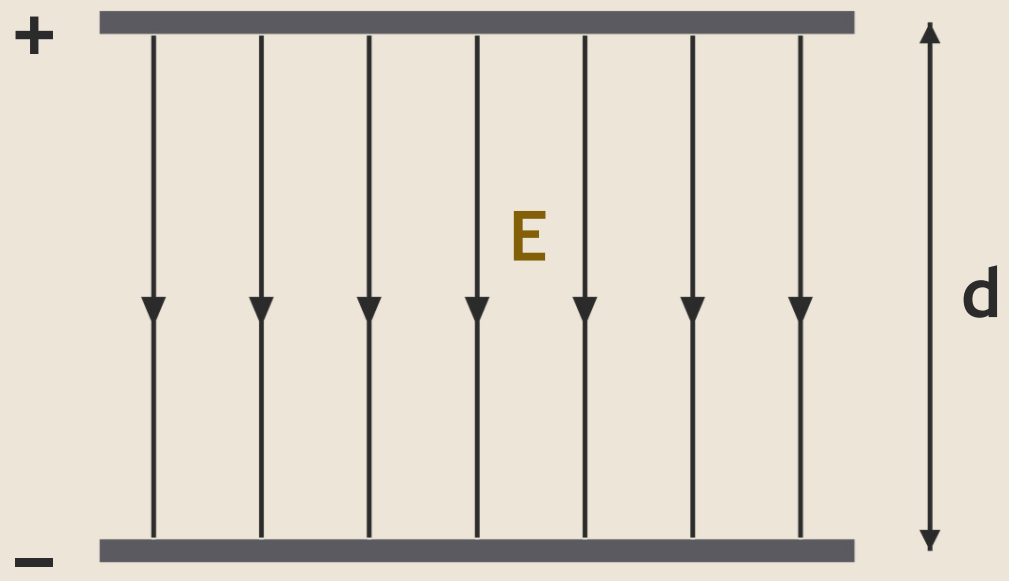
$$W = QV$$

W: work done (J)

Q: the charge (C)

V: p.d. it moves through (V)

Path does not matter (deck 1).

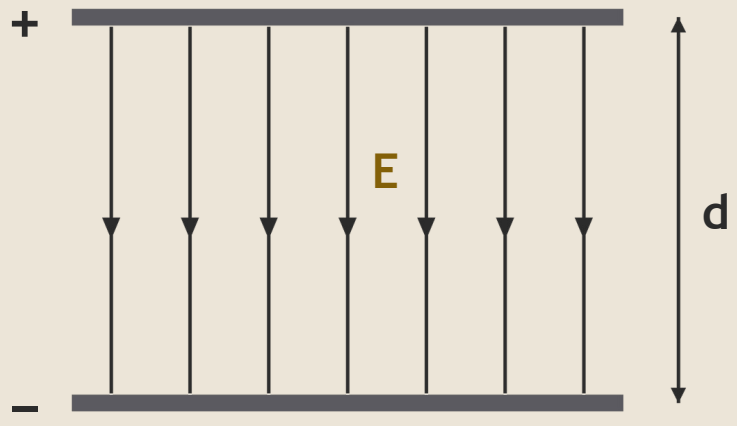


Worked example 1 — field and force

Two parallel plates are **40 mm** apart with a p.d. of **2.0 kV** across them.

(a) **Calculate** the electric field strength between the plates. **3**

(b) An electron is between the plates. **Calculate** the size of the electrical force on it. **3**



Marking: Worked example 1

Question	Expected response	Max mark	Additional guidance
(a)	$V = Ed$ (1 mark) $2.0 \times 10^{3} = E \times 0.040$ (1 mark) $E = 5.0 \times 10^{4}$ V m^{-1} (1 mark)	3	
(b)	$F = QE$ (1 mark) $F = 1.60 \times 10^{-19} \times 5.0 \times 10^{4}$ (1 mark) $F = 8.0 \times 10^{-15}$ N (1 mark)	3	

Accelerating from rest

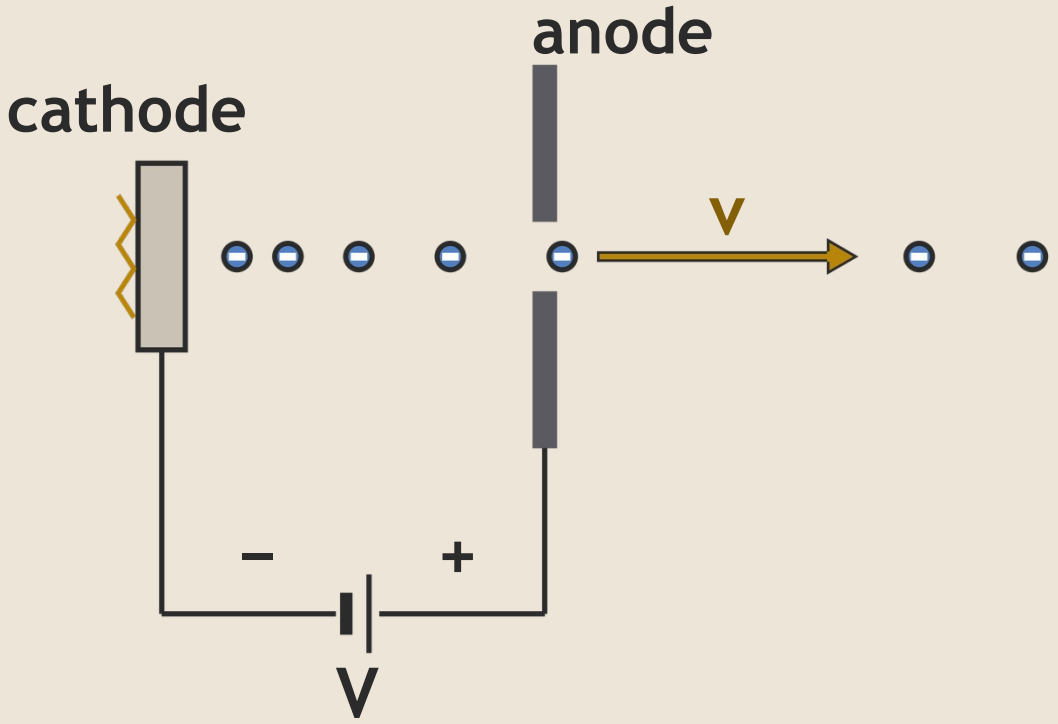
$$QV = \tfrac{1}{2}mv^2$$

Work done by the field becomes E_k.

Q, V: charge (C) and p.d. (V)

m, v: mass (kg), speed (m s^{-1})

From rest only.



How it's marked — 2022 Q14(a)

14. In a cathode ray oscilloscope, electrons are accelerated from rest between the cathode and anode. The electrons then travel with a constant horizontal velocity between the parallel deflection plates.

This arrangement is shown in Figure 14A.

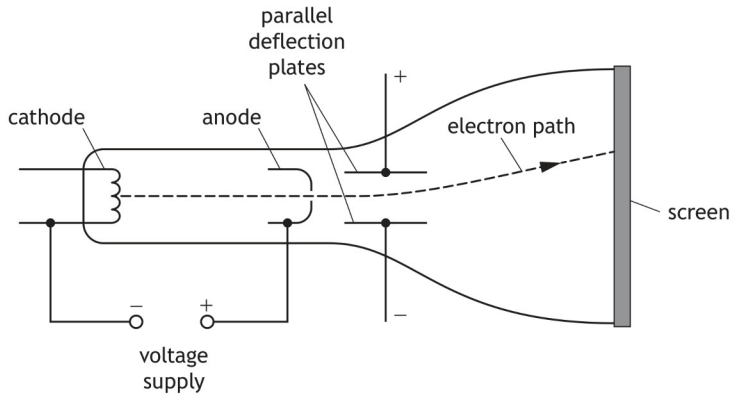


Figure 14A

(a) The electrons pass through the anode with a horizontal velocity of 2.9×10^7 m s^{-1}.

Determine the potential difference between the cathode and anode. **3**

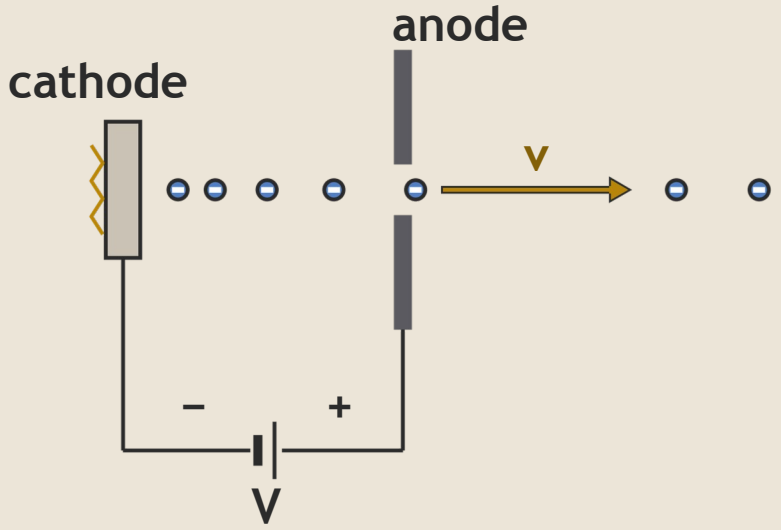
Space for working and answer

Question			Expected response	Max mark	Additional guidance
14.	(a)		$\frac{1}{2}mv^2 = QV$ **(1)**	3	Accept:2, 2.39, 2.394 Accept negative value for *Q*.
			$0.5\times9.11\times10^{-31}\times(2.9\times10^{7})^2 = 1.60\times10^{-19}\times V$ **(1)**		
			$V = 2.4\times10^{3}$ V **(1)**		

Worked example 2 — speed after accelerating

A proton is accelerated from rest through a p.d. of **2.0 kV**.

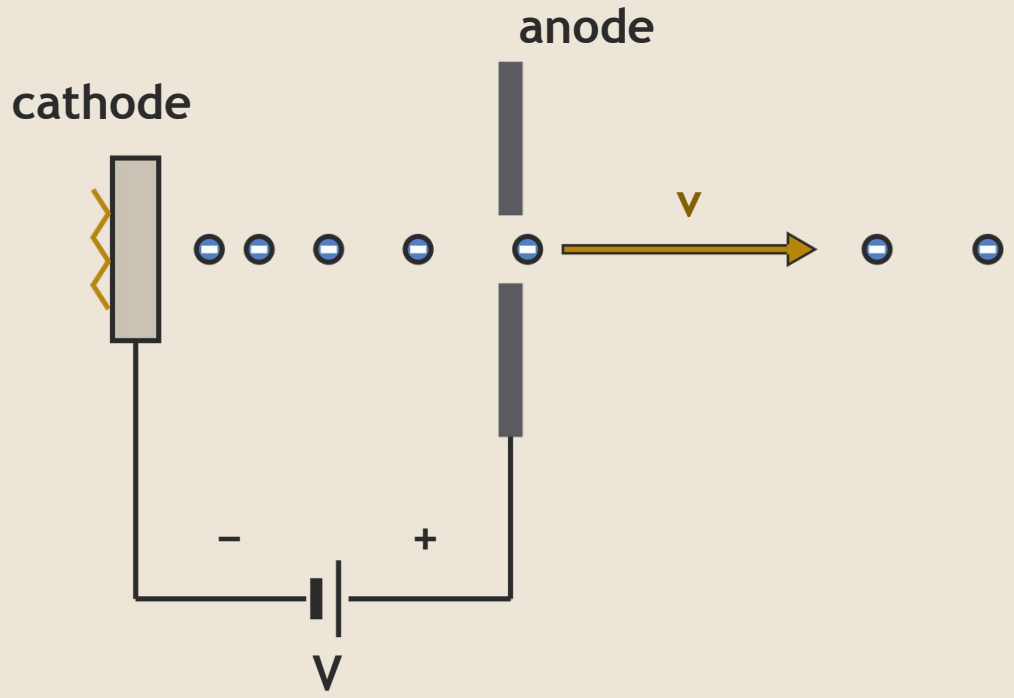
Calculate the speed of the proton. **3**



Marking: Worked example 2

Question	Expected response	Max mark	Additional guidance
	$QV = \frac{1}{2}mv^2$ (1 mark) $1.60 \times 10^{-19} \times 2.0 \times 10^{3} = \frac{1}{2} \times 1.673 \times 10^{-27} \times v^2$ (1 mark) $v = 6.2 \times 10^{5}$ m s^{-1} (1 mark)	3	Accept 6, 6.19, 6.185.

An electron is accelerated from rest through **500 V**.
Calculate its speed.



The electronvolt

LEARN THIS WORD FOR WORD

The electronvolt (eV) is the **energy acquired when one electron accelerates through a potential difference of one volt.**

Electronvolts and joules

1 eV

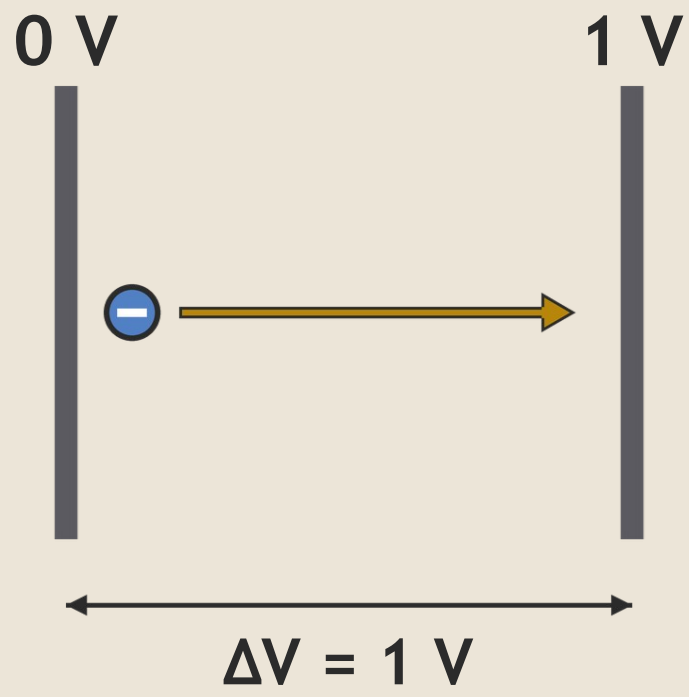
$W = QV = 1.60 \times 10^{-19}$ J.

eV to J

Multiply by 1.60×10^{-19}.

Prefixes

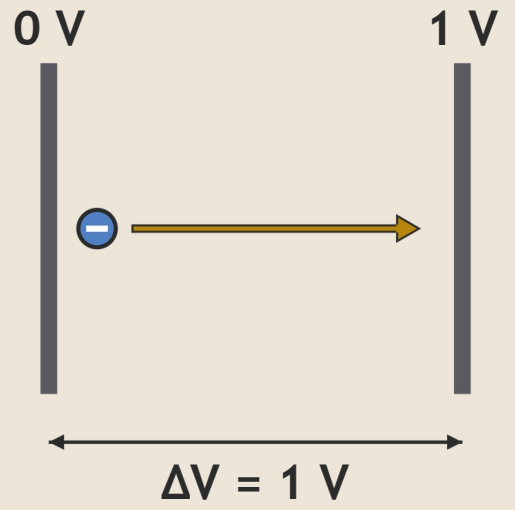
keV × 10^3, MeV × 10^6.



Worked example 3 — energy in MeV

An alpha particle of mass **6.64×10^{-27} kg** has a kinetic energy of **5.0 MeV**.

Calculate the speed of the alpha particle. 4



Marking: Worked example 3

Question	Expected response	Max mark	Additional guidance
	$E_k = \frac{1}{2}mv^2$ (1 mark) $E_k = 5.0 \times 10^6 \times 1.60 \times 10^{-19} = 8.0 \times 10^{-13}$ J (1 mark) $8.0 \times 10^{-13} = \frac{1}{2} \times 6.64 \times 10^{-27} \times v^2$ (1 mark) $v = 1.6 \times 10^7$ m s^{-1} (1 mark)	4	Independent mark for the energy conversion from MeV to J. Accept 2, 1.55, 1.552.

Exam question — 2021 Q8(c)(i)(A)

8. Polonium-212 (Po-212) undergoes nuclear decay by emitting alpha particles.

(c) Alpha particles with a specific speed are used to probe the nuclei of copper atoms in a target.

A sample of Po-212 emits alpha particles with a range of speeds.

A velocity selector is a device that will allow only alpha particles with a specific speed to pass straight through to the target.

This is shown in **Figure 8A**.

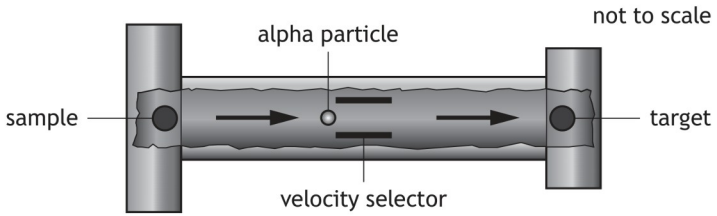


Figure 8A

(i) The velocity selector has a region in which there is a uniform electric field and a uniform magnetic field. These fields are perpendicular to each other and also perpendicular to the initial velocity v of the alpha particles, as shown in **Figure 8B**.

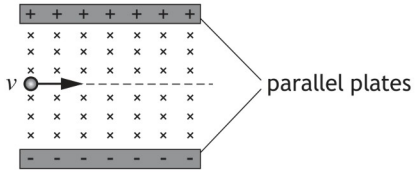


Figure 8B

(A) Calculate the speed of an alpha particle with kinetic energy 8·8 MeV. **4**

Space for working and answer

Marking — 2021 Q8(c)(i)(A)

Exam question — 2024 Q13(c)(i)

13. (a) An electric field exists around a point charge Q_1.

The charge of Q_1 is −4.5 nC.

(b) Point S is at a distance of 0.088 m from charge Q_1.

A second identical charge Q_2 is now placed at a distance of 0.088 m from point S as shown in **Figure 13A**.

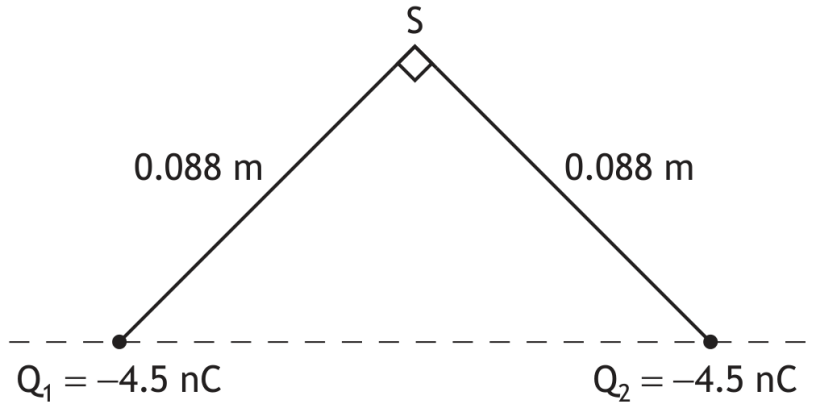


Figure 13A

(c) The electric potential at the position of Q_2, due to Q_1, is −325 V.

(i) Determine the maximum work done, in electron volts, in bringing charge Q_2 to this point. **4**

Space for working and answer

Marking — 2024 Q13(c)(i)

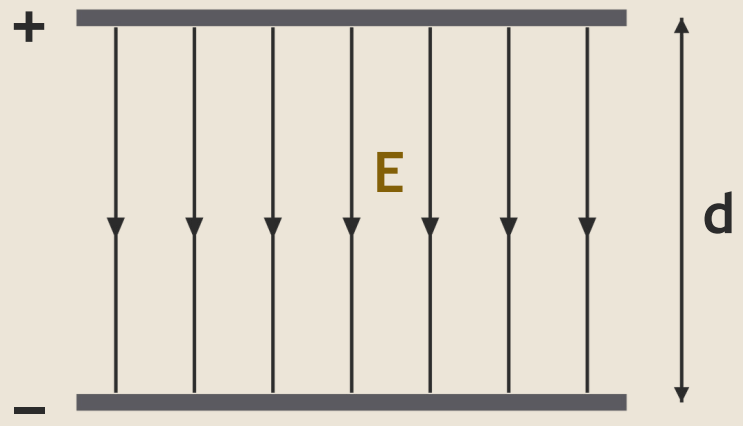
What Candidates Get Wrong

Ruler: straight, evenly spaced lines (2025 Q13(a) report).

Plate to plate, never crossing (2022 Q14(b) MI).

Arrows + to −: a mark of its own (2023 Q13(b) MI).

MeV to J: a mark of its own (2021 Q8(c)(i)(A) MI).



Look back — by the end of this part…



1. I can **sketch** the field between parallel plates.
2. I can use $E = V/d$, $F = QE$ and $W = QV$.
3. I can use $QV = \frac{1}{2}mv^2$ for a charge accelerated from rest.
4. I can **state** what the **electronvolt** is.

2025 AH Physics: scan the code and answer these.

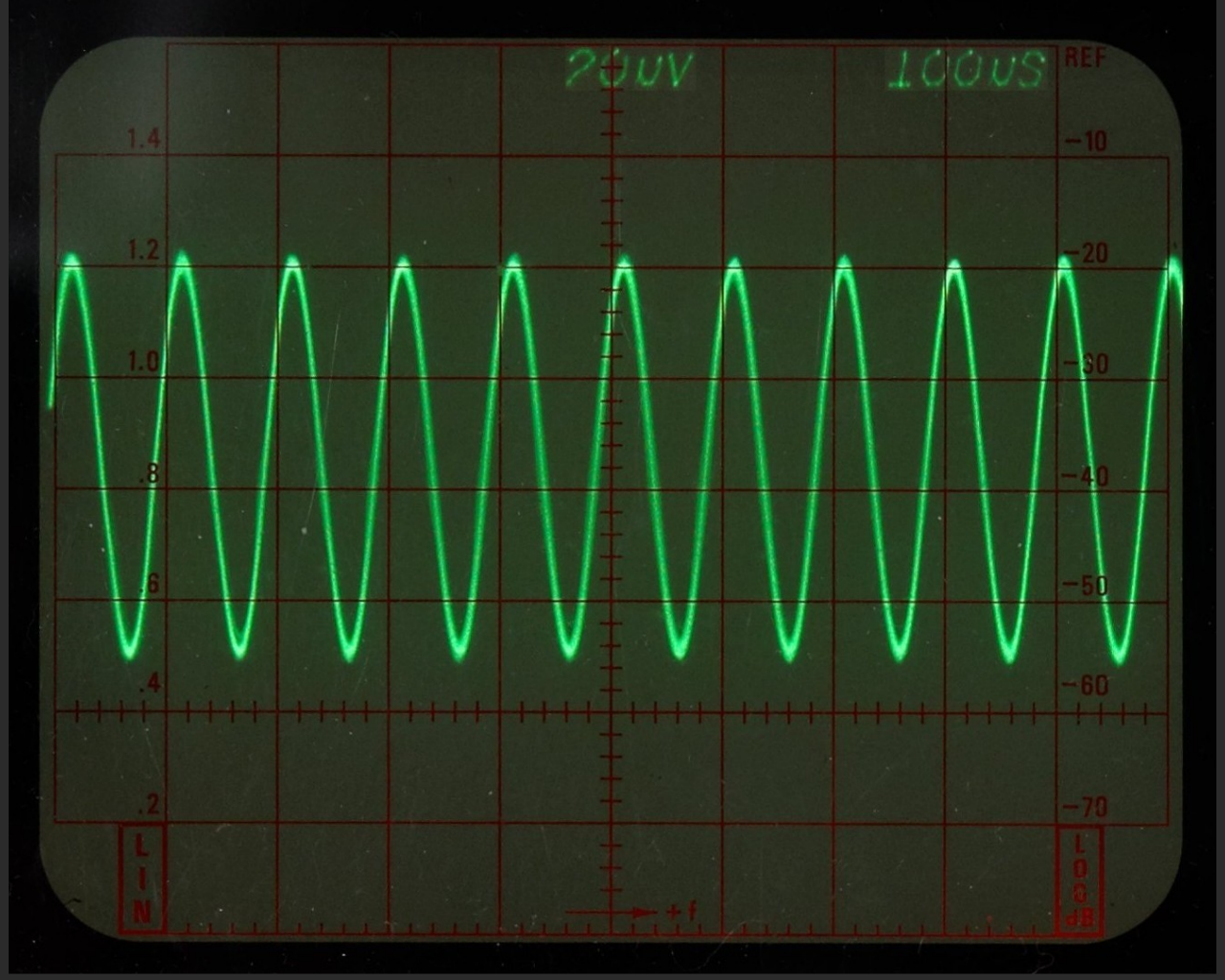
Q3(a) **Q13(b)**

Rotation, then point charges.

Show every line of working.

Scan for the paper

Marking scheme



An oscilloscope draws a wave with one beam of electrons. What bends the beam?

PART 2

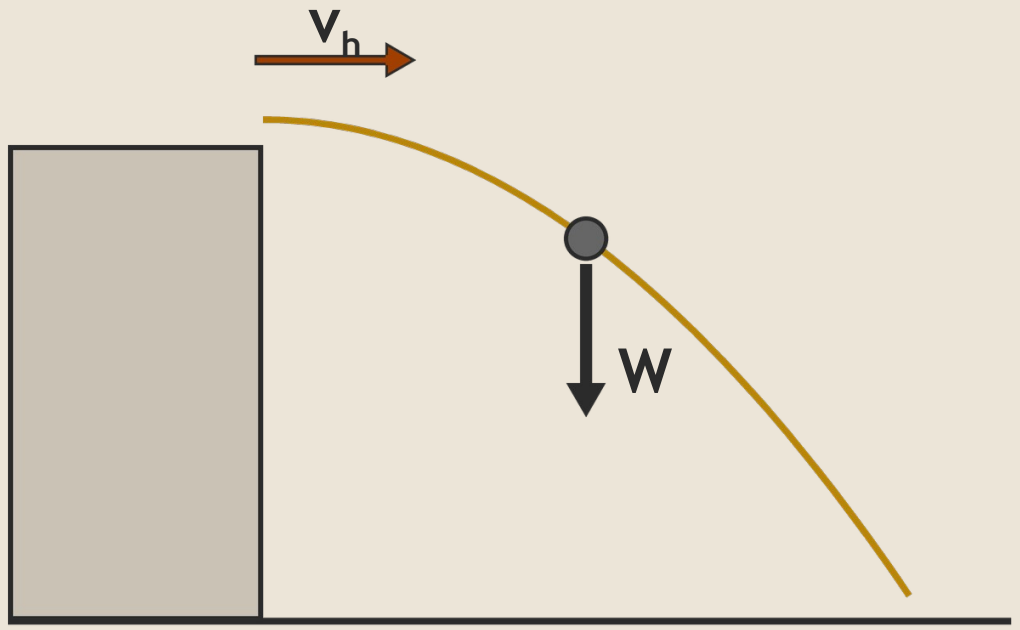
Why does the beam bend?

Between the plates an electron moves like a ball thrown sideways.

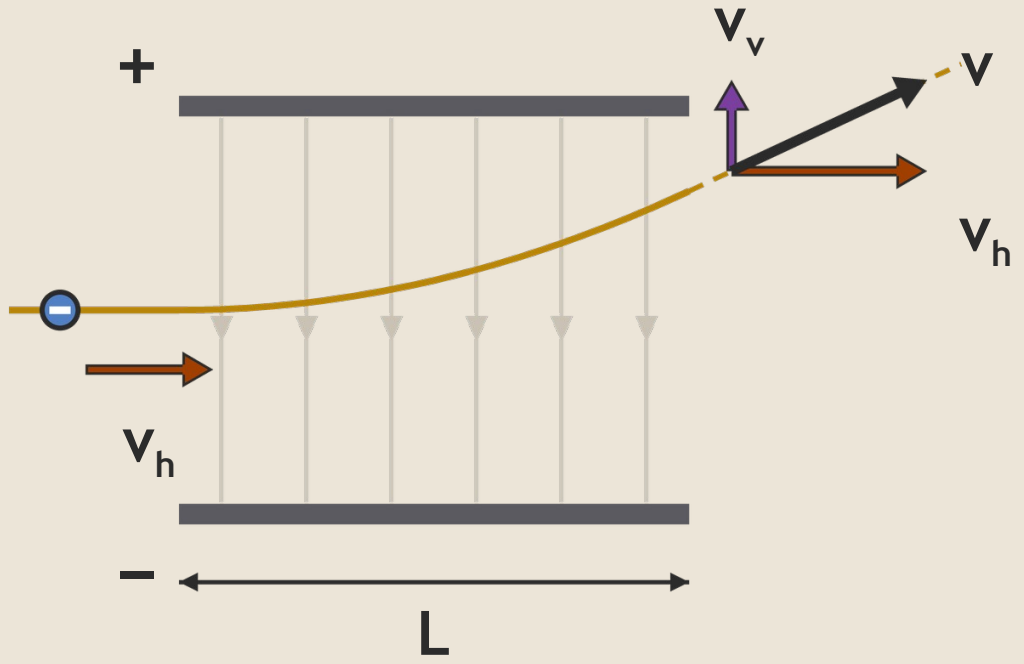
By the end of this part…

1. I can **explain** the curved path between plates.
2. I can **calculate** the time and the acceleration.
3. I can **calculate** the deflection and exit velocity.

Like a projectile — turned sideways



Gravity: curves down.



Field: curves towards +.

Two directions, two rules

Along the plates

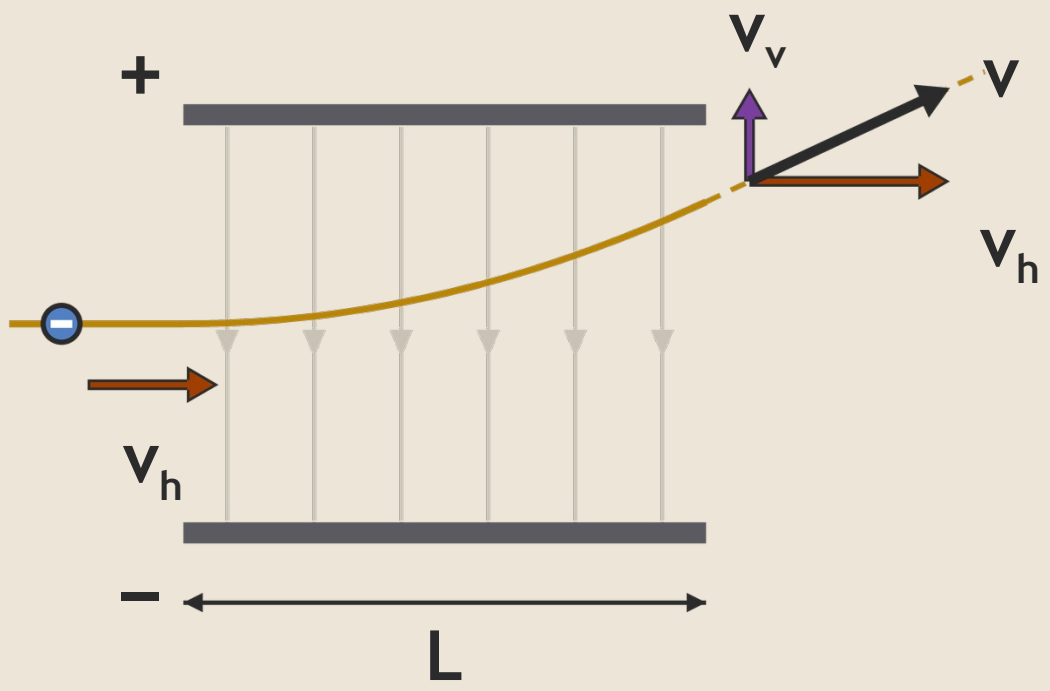
No force: v_h stays the same.

Across the plates

Force QE: steady acceleration.

Path

A parabola between the plates.



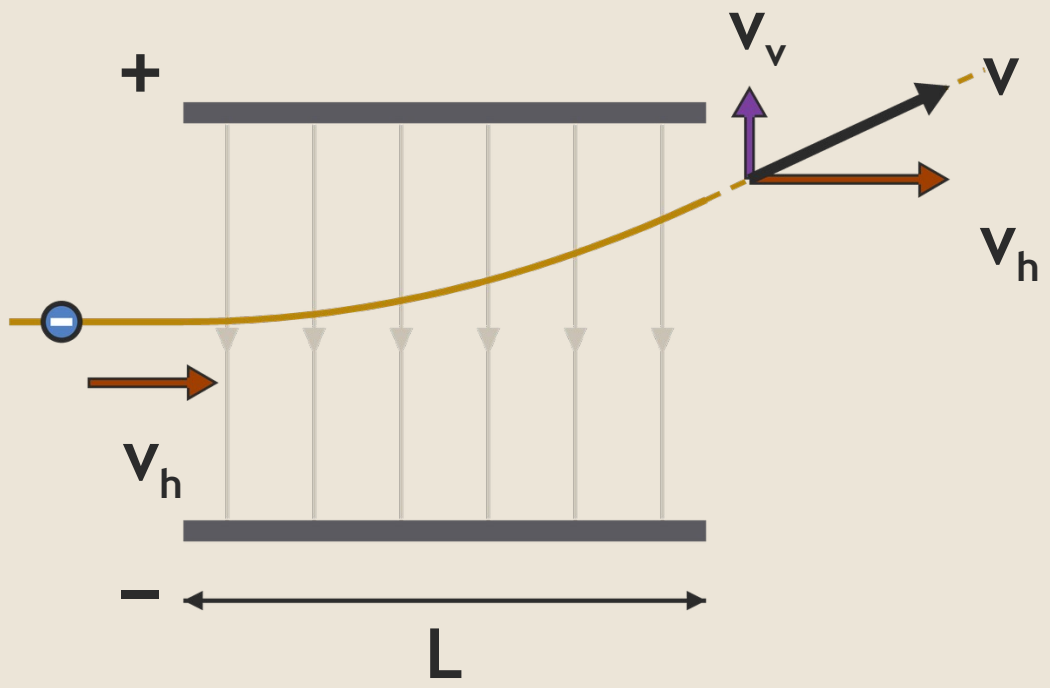
The method

$$t = L/v_h$$

$$a = QV/md$$

$$s = ut + \tfrac{1}{2}at^2$$

Across: u = 0, so s = ½at².



How it's marked — 2022 Q14(c)

14. In a cathode ray oscilloscope, electrons are accelerated from rest between the cathode and anode. The electrons then travel with a constant horizontal velocity between the parallel deflection plates.

This arrangement is shown in Figure 14A.

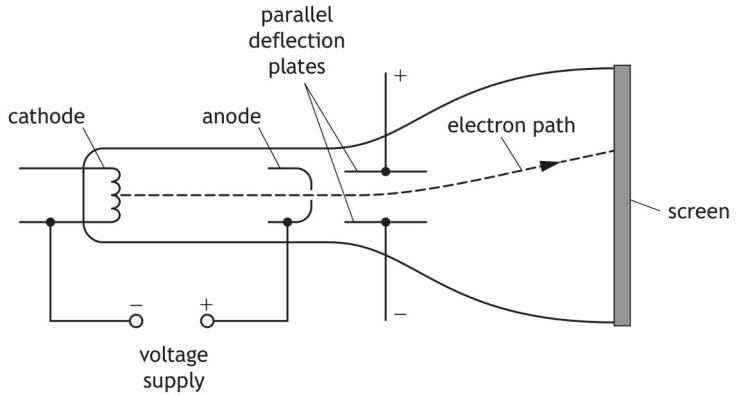


Figure 14A

(c) Explain why the electrons follow a curved path between the parallel deflection plates. **2**

[Turn over

Question			Expected response	Max mark	Additional guidance
	(c)		The electrons travel with (constant) horizontal speed/velocity. **(1)** Electrons travel with (constant) vertical acceleration. **(1)**	**2**	No force in the horizontal direction **(1)** Unbalanced force in the vertical direction **(1)** Accept: Perpendicular to field in place of horizontal Parallel to field in place of vertical. Do not accept: Attracted to the top/positive plate without reference to unbalanced force for second mark

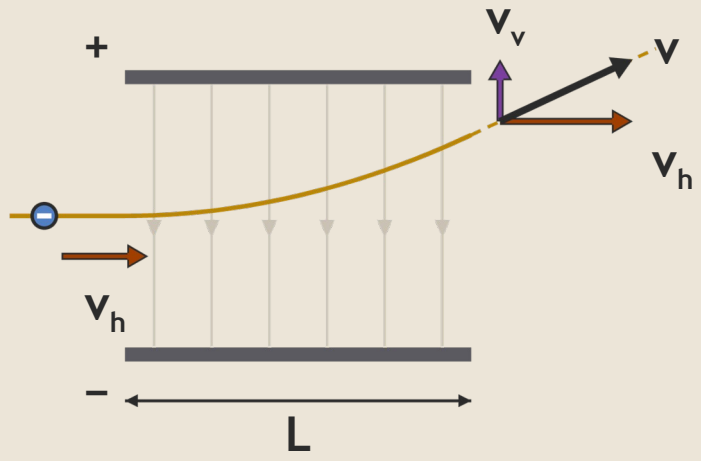
Worked example 4 — deflection

An electron enters two plates at **3.0×10^7 m s^{-1}**. The plates are **0.060 m** long, **20 mm** apart, with **150 V** across them.

(a) **Calculate** the time between the plates. **2**

(b) **Calculate** the acceleration across the plates. **3**

(c) **Calculate** the deflection as it leaves. **2**



Marking: Worked example 4 (a) and (b)

Question	Expected response	Max mark	Additional guidance
(a)	$t = L / v_h = 0.060 / 3.0 \times 10^{7}$ (1 mark) $t = 2.0 \times 10^{-9}$ s (1 mark)	2	
(b)	$F = ma$, $F = QE$ and $E = V/d$ (1 mark) $1.60 \times 10^{-19} \times 150 / 0.020 = 9.11 \times 10^{-31} \times a$ (1 mark) $a = 1.3 \times 10^{15}$ m s^{-2} (1 mark)	3	

Marking: Worked example 4 (c)

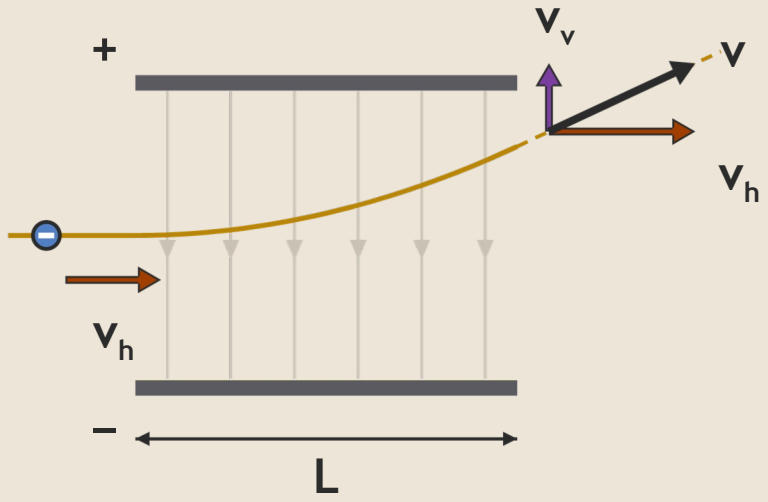
Question	Expected response	Max mark	Additional guidance
(c)	$s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 1.3 \times 10^{15} \times (2.0 \times 10^{-9})^2$ (1 mark) $s = 2.6 \times 10^{-3}$ m (1 mark)	2	

Worked example 5 — exit velocity

The electron in Worked example 4 leaves the plates.

(a) **Calculate** its vertical velocity as it leaves. **2**

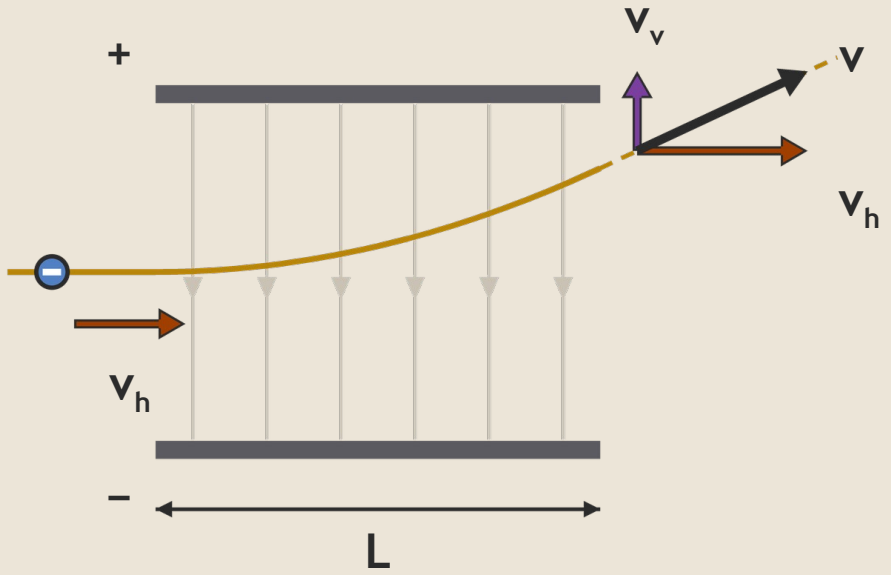
(b) **Determine** the size and direction of its velocity as it leaves. **3**



Marking: Worked example 5

Question	Expected response	Max mark	Additional guidance
(a)	$v = u + at = 0 + 1.3 \times 10^{15} \times 2.0 \times 10^{-9}$ (1 mark) $v_v = 2.6 \times 10^6$ m s^{-1} (1 mark)	2	
(b)	$v = \sqrt{((3.0 \times 10^7)^2 + (2.6 \times 10^6)^2)}$ (1 mark) $v = 3.0 \times 10^7$ m s^{-1} (1 mark) 5° above the horizontal (1 mark)	3	Trigonometry for the angle: $\tan \theta = v_v / v_h$.

An electron enters plates at **2.5×10^7 m s^{-1}**. The plates are **0.050 m** long, **15 mm** apart, with **120 V** across them. **Calculate** its deflection as it leaves.



After the plates

No field

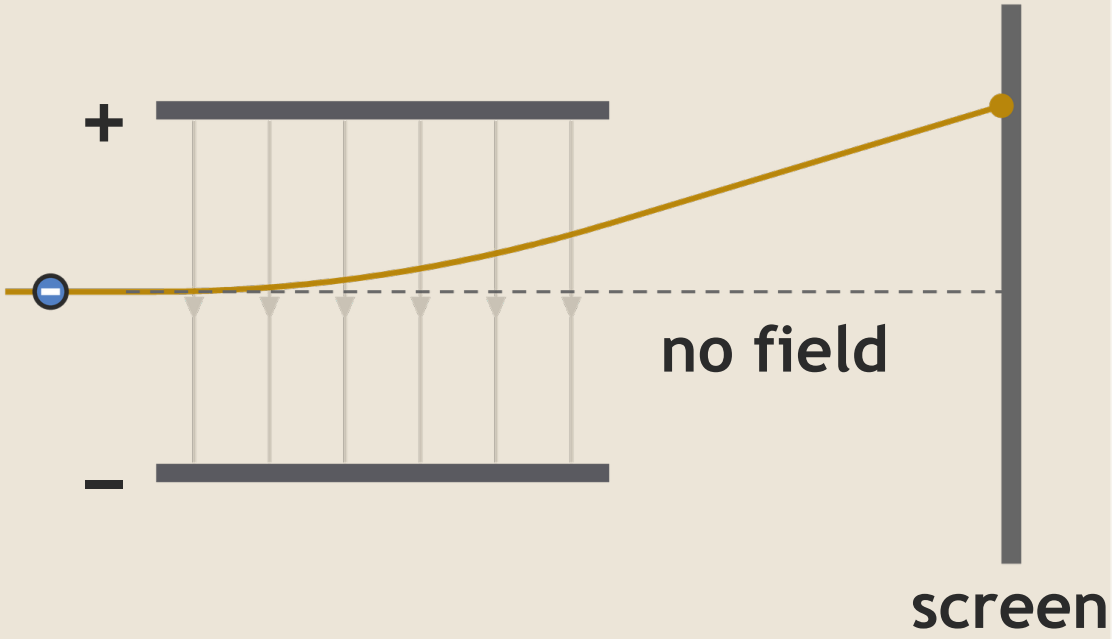
No force: a straight line.

Direction

It keeps its exit velocity.

Screen

Change V and the spot moves.



Exam question — 2022 Q14(d)

14. In a cathode ray oscilloscope, electrons are accelerated from rest between the cathode and anode. The electrons then travel with a constant horizontal velocity between the parallel deflection plates.

This arrangement is shown in Figure 14A.

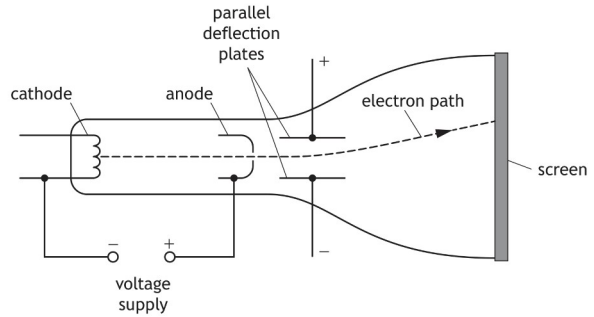


Figure 14A

(d) The potential difference across the parallel deflection plates is 0.90 kV. Electrons passing between the plates are deflected by 4.0 mm in the vertical direction.

This is shown in Figure 14B.

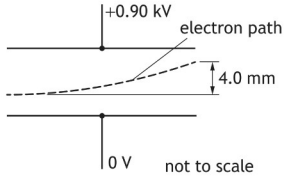


Figure 14B

(i) The vertical component of the velocity of the electrons is 1.2×10^7 m s^{-1} as they exit the region between the plates.

Show that the vertical acceleration of the electrons between the parallel deflection plates is 1.8×10^{16} m s^{-2}. **2**

Space for working and answer

(ii) By considering the electric field between the plates, determine the vertical separation of the plates. **4**

Space for working and answer

[Turn over

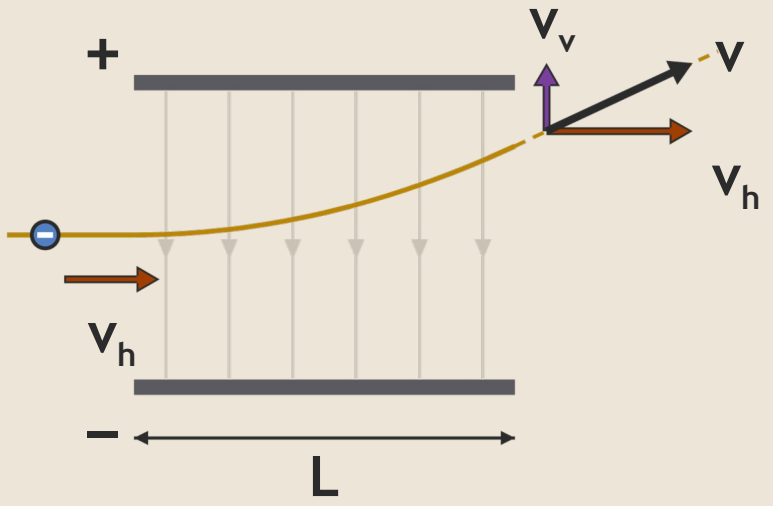
Marking — 2022 Q14(d)

What Candidates Get Wrong

Explain both: steady v_h and steady vertical acceleration (2022 Q14(c) MI).

Not enough: "attracted to the + plate" alone (2022 Q14(c) MI).

Show: final line, or 1 mark at most (2022 Q14(d)(i) MI).



Look back — by the end of this part…



1. I can **explain** the curved path between plates.
2. I can **calculate** the time and the acceleration.
3. I can **calculate** the deflection and exit velocity.

2023 AH Physics: scan the code and answer these.

Q4(a) **Q13(c)(i)**

Gravitation, then uniform fields.

Show every line of working.

Scan for the paper

Marking scheme



Millikan's apparatus, about 1916. How do you measure the charge on one electron?

PART 3

How do you measure the charge on one electron?

Millikan held a charged oil drop still in a uniform field.

By the end of this part…

1. I can **describe** Millikan's method.
2. I can use $QE = mg$ and $E = V/d$ to find the charge on a drop.
3. I can **determine** the number of electrons on a drop.

Millikan's apparatus

Oil spray

Drops are charged as they spray.

Plates

Drops fall through a hole.

Microscope

Watch one drop.

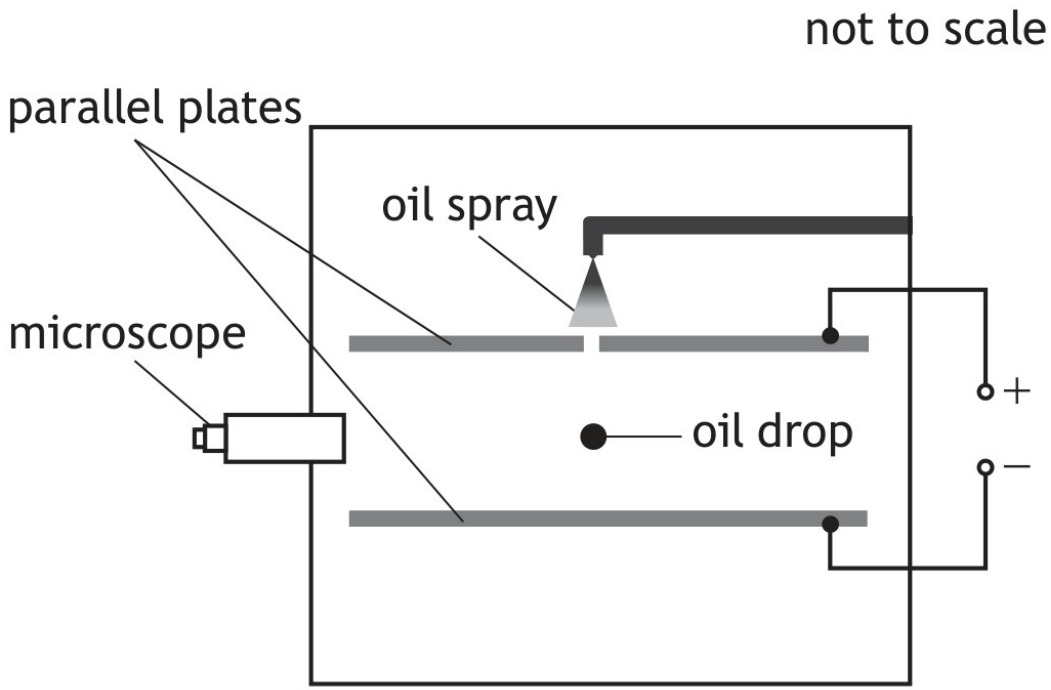


Figure 14A

Stage 1 — falling at a steady speed

Plates off

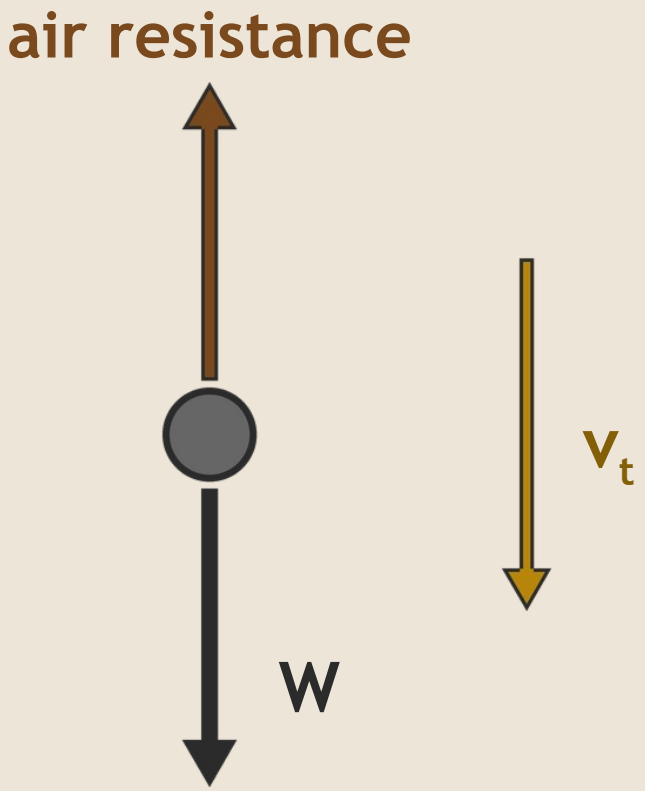
The drop falls.

Terminal velocity

Weight = air resistance.

Mass

Found from v_t. The exam gives it.



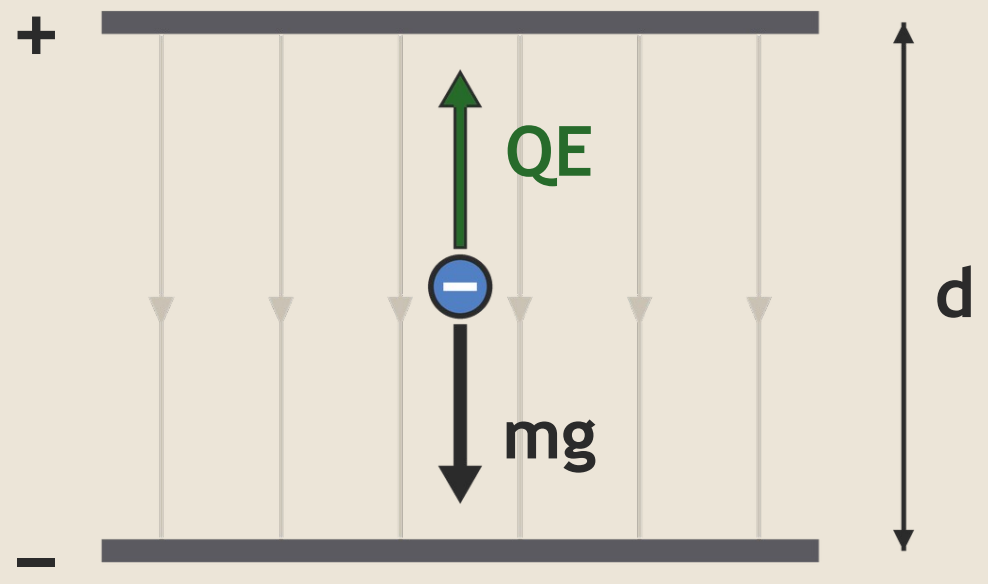
Stage 2 — held still

$$QE = mg$$

$$E = V/d$$

$$Q = mgd/V$$

Adjust V until the drop is still.



Which plate is positive?

Negative drop

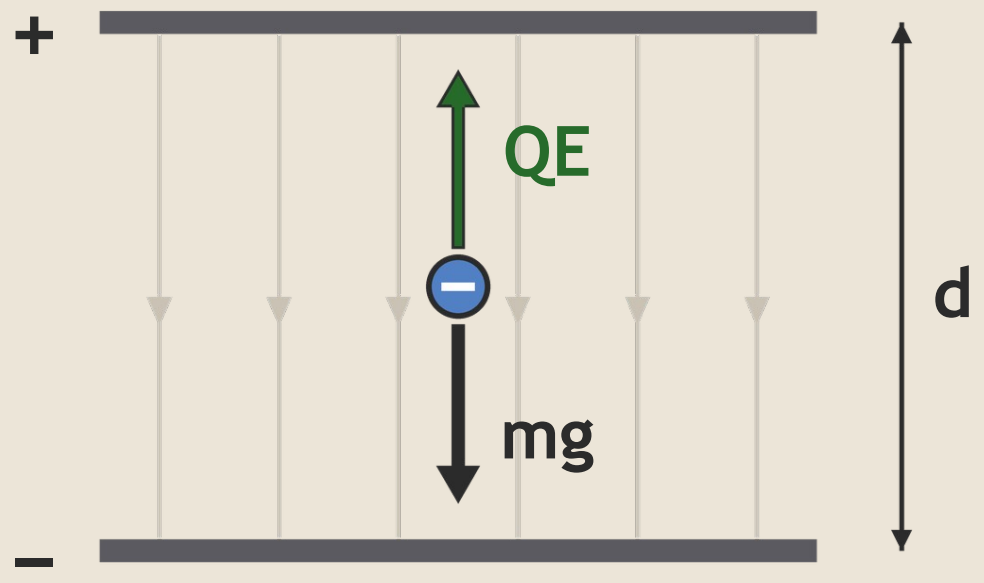
Force must point up.

Force on −

Opposite to the field.

So

The top plate is +.



The result — charge comes in steps

Every drop

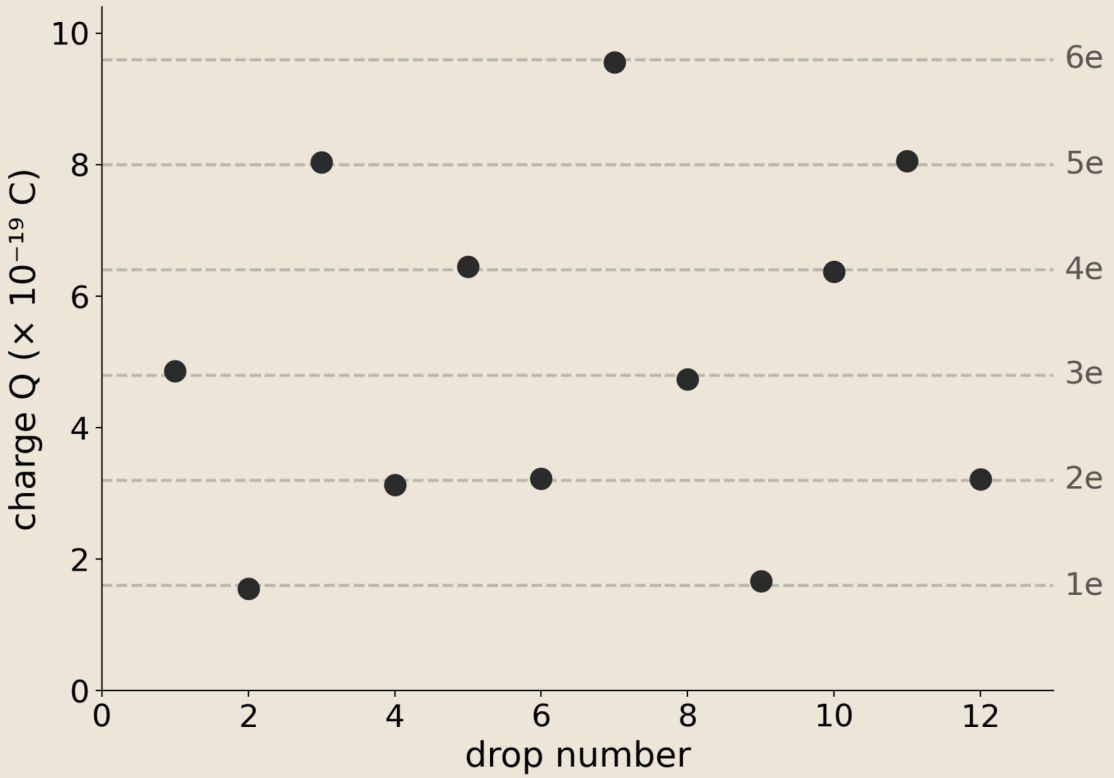
$Q = n \times 1.60 \times 10^{-19}$ C.

Quantised

n is a whole number.

e

The charge on one electron.



Illustrative values.

How it's marked — 2026 Q14(b)(ii)

14. Millikan's oil drop experiment is a method to determine the magnitude of the charge on an electron. The experiment involves observing a charged oil drop in an electric field between two parallel plates.

A simplified diagram of the apparatus is shown in **Figure 14A**.

not to scale

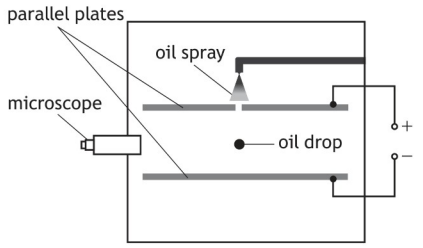


Figure 14A

The first stage of the experiment involves determining the terminal velocity v_t of the oil drop passing through the microscope field of view.

When the oil drop falls at terminal velocity, a digital stopwatch is used to measure the time for the drop to travel a set distance.

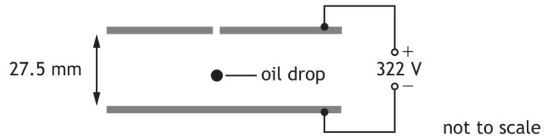
The procedure is repeated and the times recorded are

15.12 s, 13.48 s, 14.69 s, 14.63 s, 16.58 s

(b) The second stage of the experiment involves altering the potential difference between the plates until the drop is held stationary.

The drop has a mass of 9.58×10^{-16} kg and is held stationary when the potential difference between the plates is 322 V.

This is shown in **Figure 14B**.



not to scale

Figure 14B

(ii) By considering the forces acting on the drop, determine the magnitude of the charge on the drop. **3**

Space for working and answer

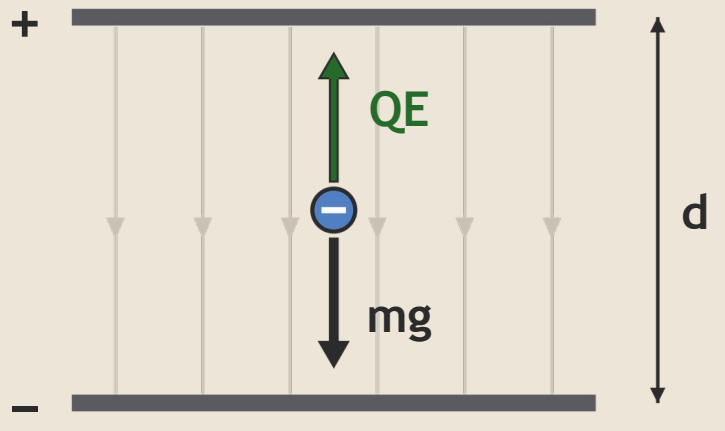
Question			Expected response	Max mark	Additional guidance
		(ii)	$F = QE$ and $W = mg$ **(1)** $Q \times 1.17 \times 10^{4} = 9.58 \times 10^{-16} \times 9.8$ **(1)** $Q = 8.0 \times 10^{-19}$ C **(1)**	3	Or consistent with (b)(i). Accept: 8, 8.02, 8.024

Worked example 6 — charge on a drop

An oil drop of mass **4.9×10^{-15} kg** is held still between two plates **15 mm** apart. The p.d. across the plates is **500 V**.

(a) By considering the forces on the drop, **determine** the charge on the drop. **3**

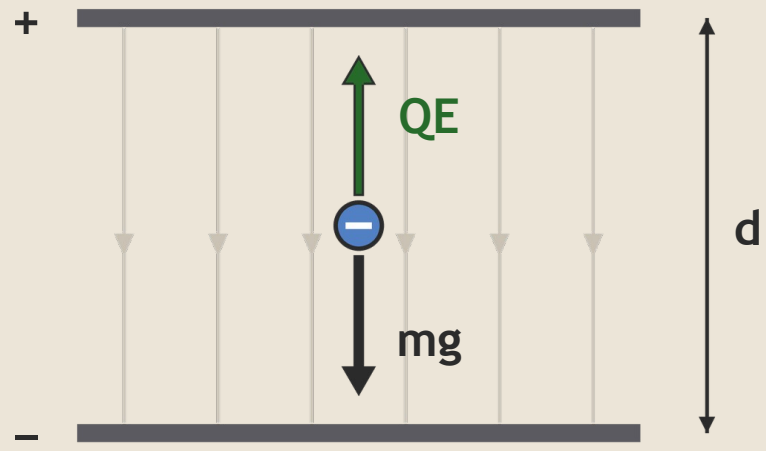
(b) **Determine** the number of excess electrons on the drop. **1**



Marking: Worked example 6

Question	Expected response	Max mark	Additional guidance
(a)	QE = mg with E = V/d (1 mark) $Q \times 500 / 0.015 = 4.9 \times 10^{-15} \times 9.8$ (1 mark) $Q = 1.4 \times 10^{-18}$ C (1 mark)	3	
(b)	$n = 1.4 \times 10^{-18} / 1.60 \times 10^{-19} = 9$ (1 mark)	1	Must be an integer.

A drop of mass **3.2 × 10^{-15} kg** is held still by **980 V** across plates **20 mm** apart. **Determine** the charge on it and the number of excess electrons.



Exam question — 2026 Q14(b)(i)

14. Millikan's oil drop experiment is a method to determine the magnitude of the charge on an electron. The experiment involves observing a charged oil drop in an electric field between two parallel plates.

A simplified diagram of the apparatus is shown in **Figure 14A**.

not to scale

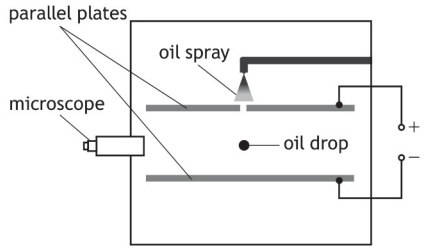


Figure 14A

The first stage of the experiment involves determining the terminal velocity v_t of the oil drop passing through the microscope field of view.

When the oil drop falls at terminal velocity, a digital stopwatch is used to measure the time for the drop to travel a set distance.

The procedure is repeated and the times recorded are

15.12 s, 13.48 s, 14.69 s, 14.63 s, 16.58 s

(b) The second stage of the experiment involves altering the potential difference between the plates until the drop is held stationary.

The drop has a mass of 9.58×10^{-16} kg and is held stationary when the potential difference between the plates is 322 V.

This is shown in **Figure 14B**.

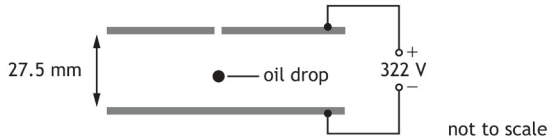


Figure 14B

(i) Calculate the electric field strength between the plates. **3**

Space for working and answer

Marking — 2026 Q14(b)(i)

Exam question — 2026 Q14(b)(iii)

14. Millikan's oil drop experiment is a method to determine the magnitude of the charge on an electron. The experiment involves observing a charged oil drop in an electric field between two parallel plates.

A simplified diagram of the apparatus is shown in **Figure 14A**.

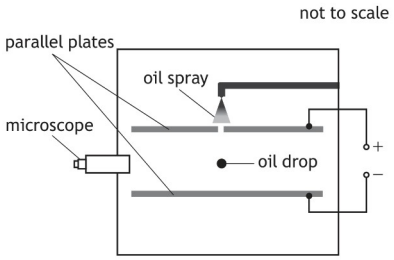


Figure 14A

The first stage of the experiment involves determining the terminal velocity v_t of the oil drop passing through the microscope field of view.

When the oil drop falls at terminal velocity, a digital stopwatch is used to measure the time for the drop to travel a set distance.

The procedure is repeated and the times recorded are

15.12 s, 13.48 s, 14.69 s, 14.63 s, 16.58 s

(b) The second stage of the experiment involves altering the potential difference between the plates until the drop is held stationary.

The drop has a mass of 9.58×10^{-16} kg and is held stationary when the potential difference between the plates is 322 V.

This is shown in **Figure 14B**.

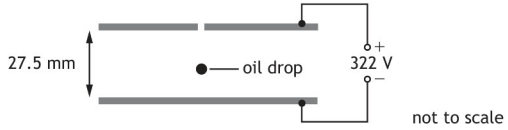


Figure 14B

(ii) By considering the forces acting on the drop, determine the magnitude of the charge on the drop. **3**

Space for working and answer

(iii) Determine the number of excess electrons on the drop. **1**

Space for working and answer

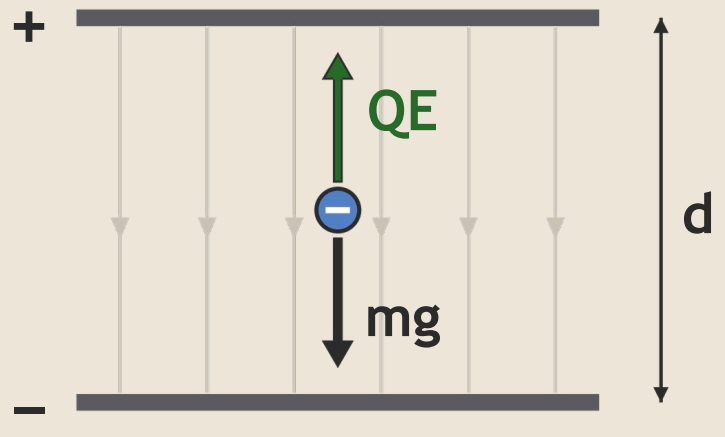
Marking — 2026 Q14(b)(iii)

What Candidates Get Wrong

Forces first: write QE = mg (2026 Q14(b)(ii): 48 %).

Whole number of electrons (2026 Q14(b)(iii): 43 %).

Metres: 27.5 mm = 0.0275 m (2026 Q14(b)(i) MI).



Look back — by the end of this part…



1. I can **describe** Millikan's method.
2. I can use $QE = mg$ and $E = V/d$ to find the charge on a drop.
3. I can **determine** the number of electrons on a drop.

Uniform fields in the exam

Year	Question	Marks
2026	Q14 (b)(i) to (iii)	7
2025	Q14 (a)(i) to (ii) *	5
2024	Q13 (c)(i)	4
2023	Q13 (c)(i)	3
2022	Q14 (a) to (d)	12
2021	Q8 (c)(i)(A)	4

* with a magnetic field (a later topic)

Past paper finder

Reading and homework

Scholar Topic 3

Scholar — Unit 3, Topic 3 Motion in an electric field
BrightRed Study Guide pages 96-97
Past papers — the questions on the last slide

Scholar

Errors that cost marks

- Field lines: ruler, evenly spaced, plate to plate, arrows + to −.
- MeV to J before $\frac{1}{2}mv^2$: the conversion is a mark.
- Curved path: constant v_h **and** constant vertical acceleration.
- Millikan: write QE = mg first; electrons are a whole number.
- Change mm to m and kV to V first.

Check yourself

1. **Sketch** the field between two parallel plates.
2. **Calculate** E, F and W between plates.
3. **Calculate** a speed after accelerating through a p.d.
4. **Explain** the curved path, then **calculate** the deflection.
5. **Describe** Millikan's method and find the charge on a drop.

Experiments you could try

- A Millikan oil drop simulation: balance a drop, find its charge.
- PhET "Charges and Fields": a row of + above a row of − makes a near-uniform field.
- Watch a Teltron deflection tube video: change V, see the beam bend.
- Research: how an old CRT TV or oscilloscope steers its beam.

Course specification — 1 of 3

- Sketch of electric field patterns around single point charges, a system of charges and in a uniform electric field.
- Use of appropriate relationships to solve problems involving charge, energy, potential difference, and electric field strength in situations involving a uniform electric field.
- $F = QE$ $V = Ed$ $W = QV$



Course spec, pages 17-18

Course specification — 2 of 3

- Knowledge of Millikan's experimental method for determining the charge on an electron.
- Use of appropriate relationships to solve problems involving the motion of charged particles in uniform electric fields.
- $F = QE \quad V = Ed \quad W = QV \quad E_k = \frac{1}{2}mv^2$

Course spec, pages 17-18

Course specification — 3 of 3

- Knowledge that the electronvolt (eV) is the energy acquired when one electron accelerates through a potential difference of one volt.
- Conversion between electronvolts and joules.



Course spec, pages 17-18

Credits (1 of 2)

Starter: 2024 AH Physics, © Qualifications Scotland (SQA); 2025 AH Physics, © Qualifications Scotland (SQA); 2023 AH Physics, © Qualifications Scotland (SQA)

Part 1 hook: Photo: Mister rf, CC BY-SA 4.0, Wikimedia Commons

Example answer — 2022 Q14(b): © Qualifications Scotland (SQA). AH Physics 2022, Q14(b)

Marking: 2022 Q14(b): © Qualifications Scotland (SQA). AH Physics 2022 marking instructions, Q14(b)

Exam rounds and how it's marked: © Qualifications Scotland (SQA). AH Physics 2022 Q14; © Qualifications Scotland (SQA). AH Physics 2021 Q8; © Qualifications Scotland (SQA). AH Physics 2024 Q13; © Qualifications Scotland (SQA). AH Physics 2026 Q14

The electronvolt: © Qualifications Scotland (SQA)

Part 2 hook: Photo: Pittigrilli, CC BY-SA 4.0, Wikimedia Commons

Part 3 hook: Photo: Daderot, CC0, Wikimedia Commons

Credits (2 of 2)

Millikan's apparatus: © Qualifications Scotland (SQA). AH Physics 2026, Q14 Figure 14A

Uniform fields in the exam: © Qualifications Scotland (SQA). AH Physics 2021-2026.

Reading and homework: SCHOLAR AH Physics © Heriot-Watt University. BrightRed AH Physics Study Guide

Course specification — 1 of 3: © Qualifications Scotland (SQA). AH Physics course specification v3.1, pages 17-18

Course specification — 2 of 3: © Qualifications Scotland (SQA). AH Physics course specification v3.1, pages 17-18

Course specification — 3 of 3: © Qualifications Scotland (SQA). AH Physics course specification v3.1, pages 17-18

R Stewart - November 2026